Homework 3

Background

The national ambient air quality standard for fine particulate matter, or airborne particles that are smaller than 2.5 μm in diameter (PM2.5) is 35 μg/m^3. Daily concentrations of PM2.5 cannot exceed this level when averaged over a 24-hour period. In many cities across the U.S. we see PM2.5 levels exceed the 35 μg/m^3 standard and one question is whether levels above the standard are harmful to human health. In the past, many have argued that 35 was too low of a level and that exposure to PM2.5 was safe at a higher level.

In this homework is a dataset for New York City on daily mortality and PM2.5. Also included are daily temperature (in degrees F) and an indicator of whether the level of PM2.5 was above 35 on a given day ("above" or "below"). Here are a random few rows of the dataset.

```
# A tibble: 10 × 5
   date       death  temp  pm25 above
   <date>     <dbl> <dbl> <dbl> <chr>
 1 1999-04-28   153  54   NA    <NA>
 2 2000-06-17   154  80.7 19.7  below
 3 2000-07-19   122  66.3 10.2  below
 4 2000-08-03   150  76.3 22.7  below
 5 2001-08-25   144  72.3  7.7  below
 6 2002-02-05   202  26.7  6.85 below
 7 2003-02-16   166  12.7  8.65 below
 8 2003-03-31   169  37.3  5.16 below
 9 2003-11-08   157  43    5.79 below
10 2005-07-01   143  75.7 15.6  below
```

Each row in the dataset represents a day. The `death` column shows the total number of non-accidental deaths on that day, the `temp` column shows the average temperature for that day, the `pm25` shows the PM2.5 level for that day, and the `above` column indicates whether PM2.5 on that day was above 35.

Analysis

Your task in this homework is to use propensity score matching methods to answer the question: **Do PM2.5 levels above the 35 μg/m^3 standard cause excess daily mortality in New York City?** Excess daily mortality is mortality greater than what might be expected on a day in New York. For this analysis, you must:

1. Draw a directed acyclic graph of the relevant causal relationships for your analysis. Consider what might be a potential confounder of the PM2.5 above/below –> mortality relationship and see if there is a way to account for it using the data that you have.
2. Build a propensity score model using the confounders specified in your DAG. You can use the `MatchIt` package for this if you want (or you can code it from scratch).
3. Use the propensity score model to build a matched dataset and estimate the causal effect of having PM2.5 levels above the 35 μg/m^3 standard.
4. Identify two decisions that you made in your analysis that an outside observer might want to change. Generate an alternative scenario that this hypothetical outside observer might want to see and conduct an analysis of the data using this alternative scenario.

Answer the following questions in your analysis:

1. Do PM2.5 levels above the 35 μg/m^3 standard cause excess daily mortality in New York City?
2. How would you rate the strength of the evidence in this analysis?

Note that there are relatively few variables in this dataset. However, consider what could be done with the `date` variable in the context of this analysis, where the unit of analysis is a day.

Report

Write up your analysis in an R markdown or Quarto document so that you can include all code used for the analysis along with results. Compile your document to PDF and submit it to Canvas.